



Week 3 Status Report

Project: CCP-CN-S2026 – Problem 10: High Availability Router

Team Name: Guardian Gateways

Week2: (GNS3 Implementation & Basic Connectivity)

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Scenario:

A water treatment plant cannot afford downtime. Two routers must provide gateway redundancy: if Router A fails, Router B takes over automatically within a few seconds.

Devices Used:

- 2 Cisco 4331 Routers
- 1 Core Switch
- 2 VPCS

Introduction:

Implemented security controls on High Availability Router network.

Security Features Implemented:

ACL

SSH

Login Blocking

MOTD Banner

HSRP Verification

Interface Tracking

Testing Performed

- ACL test
- SSH test
- Failover test
- Recovery test
- Connectivity test

Results

- Unauthorized access blocked
- SSH working
- HSRP failover successful
- Preemption successful
- Network remained available

Week 3 Screenshots List:

Standard ACL:

```
RouterA(config)#access-list 10 permit 192.168.10.50
RouterA(config)#access-list 10 deny any
RouterA(config)#line vty 0 4
RouterA(config-line)#access-class 10 in
RouterA(config-line)#transport input ssh
RouterA(config-line)#login local
RouterA(config-line)#show access-lists
      ^
% Invalid input detected at '^' marker.

RouterA(config-line)#end
RouterA#show access-lists
*Jun 13 23:15:47.219: %SYS-5-CONFIG_I: Configured from console by console
Standard IP access list 10
  10 permit 192.168.10.50
  20 deny any
RouterA#conf t
Enter configuration commands, one per line. End with CNTL/Z.
RouterA(config)#no ip http server
RouterA(config)#no ip http secure-server
RouterA(config)#service password-encryption
RouterA(config)#
```

ACL CONFIGURATION:

```
RouterA#conf t
Enter configuration commands, one per line. End with CNTL/Z.
RouterA(config)#access-list 101 deny ip host 192.168.10.50 any
RouterA(config)#access-list 101 permit ip any any
RouterA(config)#interface g1/0
RouterA(config-if)#ip access-group 101 in
RouterA(config-if)#exit
RouterA(config)#end
RouterA#
```

ACL hit counts:

```
RouterA#show access-list
Standard IP access list 10
  10 permit 192.168.10.50
  20 deny any
Extended IP access list 101
  10 deny ip host 192.168.10.50 any
  20 permit ip any any (13 matches)
```

ACL was implemented to restrict unauthorized access. Hit counts confirm that the ACL is actively filtering traffic.

Unauthorized PC Test:

```
84 bytes from 192.168.10.10: icmp_seq=3 ttl=255 time=15.316 ms

VPCS> ip 192.168.20.10/24 192.168.20.1ip 192.168.20.10/24 192.168.20.1
Checking for duplicate address...
PC1 : 192.168.20.10 255.255.255.0 gateway 192.168.20.1

VPCS> ping 192.168.10.10
host (192.168.20.1) not reachable
```

Secure SSH Only:

```
RouterA#show running-config | section vty
line vty 0 4
  access-class 10 in
  login local
  transport input ssh
RouterA#show login
  No login delay has been applied.
  No Quiet-Mode access list has been configured.

  Router NOT enabled to watch for login Attacks
```

Attack Execution Script & Logs:

```
VPCS> ping 192.168.10.2 -f
84 bytes from 192.168.10.2 icmp_seq=1 ttl=255 time=15.339 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=255 time=15.336 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=255 time=15.316 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=255 time=16.239 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=255 time=19.969 ms
```

```
RouterA#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited, 0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

  Console logging: level debugging, 44 messages logged, xml disabled,
                    filtering disabled
  Monitor logging: level debugging, 0 messages logged, xml disabled,
                    filtering disabled
  Buffer logging: level debugging, 44 messages logged, xml disabled,
                  filtering disabled
  Exception Logging: size (8192 bytes)
  Count and timestamp logging messages: disabled
  Persistent logging: disabled

No active filter modules.
```

A controlled traffic flood was generated to test network resilience and security controls.

ACL hit counts increasing

```
RouterA#show access-list
Standard IP access list 10
 10 permit 192.168.10.50
 20 deny any
Extended IP access list 101
 10 deny ip host 192.168.10.50 any
 20 permit ip any any log (330 matches)
```

ACL-based filtering was applied to mitigate excessive ICMP traffic.

Login Block Feature

```
RouterA#show login
No login delay has been applied.
No Quiet-Mode access list has been configured.

Router NOT enabled to watch for login Attacks

RouterA#show standby brief
P indicates configured to preempt.
|
Interface Grp Pri P State Active Standby Virtual IP
Gig0/0 1 120 P Active local 192.168.10.3 192.168.10.1
RouterA#
```

Banner Configure:

```
RouterA#conf t
Enter configuration commands, one per line. End with CNTL/Z.
RouterA(config)#banner motd #
Enter TEXT message. End with the character '#'.
WARNING!
Authorized Users Only.Unauthorized Access Prohibited.
#
RouterA(config)#conf t
```

Save Configuration:

```

*Jun 13 23:27:44.423: %SYS-5-CONFIG_I: Configured from console by console
RouterA#show running-config
Building configuration...

Current configuration : 1224 bytes
!
! Last configuration change at 23:27:44 UTC Sat Jun 13 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname RouterA
!
boot-start-marker
boot-end-marker
!
!
enable secret 5 $1$usQH$4FiQokKEIpMnfZt4ct5xp.
!
no aaa new-model
ip cef
!
!
!

```

Verify HSRP Still Working

```

*Jun 13 23:35:39.911: %SYS-5-CONFIG_I: Configured from console by console
RouterA#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri P State  Active      Standby      Virtual IP
Gi1/0      1    120 P Active local     192.168.10.3 192.168.10.1
RouterA#

```

Continuous Ping Test:

```

VPCS> ping 192.168.10.20 -t
84 bytes from 192.168.10.20 icmp_seq=1 ttl=64 time=0.702 ms
84 bytes from 192.168.10.20 icmp_seq=2 ttl=64 time=0.894 ms
84 bytes from 192.168.10.20 icmp_seq=3 ttl=64 time=59.594 ms
84 bytes from 192.168.10.20 icmp_seq=4 ttl=64 time=0.649 ms
84 bytes from 192.168.10.20 icmp_seq=5 ttl=64 time=0.648 ms
84 bytes from 192.168.10.20 icmp_seq=6 ttl=64 time=0.745 ms
84 bytes from 192.168.10.20 icmp_seq=7 ttl=64 time=0.678 ms
84 bytes from 192.168.10.20 icmp_seq=8 ttl=64 time=37.179 ms
84 bytes from 192.168.10.20 icmp_seq=9 ttl=64 time=0.762 ms
84 bytes from 192.168.10.20 icmp_seq=10 ttl=64 time=0.854 ms
84 bytes from 192.168.10.20 icmp_seq=11 ttl=64 time=0.690 ms
84 bytes from 192.168.10.20 icmp_seq=12 ttl=64 time=106.036 ms
84 bytes from 192.168.10.20 icmp_seq=13 ttl=64 time=1.095 ms
84 bytes from 192.168.10.20 icmp_seq=14 ttl=64 time=1.436 ms
84 bytes from 192.168.10.20 icmp_seq=15 ttl=64 time=0.664 ms
84 bytes from 192.168.10.20 icmp_seq=16 ttl=64 time=1.137 ms
84 bytes from 192.168.10.20 icmp_seq=17 ttl=64 time=0.631 ms
84 bytes from 192.168.10.20 icmp_seq=18 ttl=64 time=0.655 ms
84 bytes from 192.168.10.20 icmp_seq=19 ttl=64 time=0.707 ms
84 bytes from 192.168.10.20 icmp_seq=20 ttl=64 time=0.671 ms
84 bytes from 192.168.10.20 icmp_seq=21 ttl=64 time=0.639 ms
84 bytes from 192.168.10.20 icmp_seq=22 ttl=64 time=1.218 ms
84 bytes from 192.168.10.20 icmp_seq=23 ttl=64 time=0.828 ms
84 bytes from 192.168.10.20 icmp_seq=24 ttl=64 time=1.704 ms
84 bytes from 192.168.10.20 icmp_seq=25 ttl=64 time=0.663 ms
84 bytes from 192.168.10.20 icmp_seq=26 ttl=64 time=0.665 ms
84 bytes from 192.168.10.20 icmp_seq=27 ttl=64 time=0.632 ms
84 bytes from 192.168.10.20 icmp_seq=28 ttl=64 time=0.632 ms
84 bytes from 192.168.10.20 icmp_seq=29 ttl=64 time=0.698 ms
84 bytes from 192.168.10.20 icmp_seq=30 ttl=64 time=0.713 ms
84 bytes from 192.168.10.20 icmp_seq=31 ttl=64 time=0.619 ms
84 bytes from 192.168.10.20 icmp_seq=32 ttl=64 time=0.772 ms
84 bytes from 192.168.10.20 icmp_seq=33 ttl=64 time=6.934 ms
84 bytes from 192.168.10.20 icmp_seq=34 ttl=64 time=0.886 ms
84 bytes from 192.168.10.20 icmp_seq=35 ttl=64 time=0.638 ms
84 bytes from 192.168.10.20 icmp_seq=36 ttl=64 time=0.636 ms
84 bytes from 192.168.10.20 icmp_seq=37 ttl=64 time=0.637 ms
84 bytes from 192.168.10.20 icmp_seq=38 ttl=64 time=2.109 ms
84 bytes from 192.168.10.20 icmp_seq=39 ttl=64 time=0.625 ms
84 bytes from 192.168.10.20 icmp_seq=40 ttl=64 time=0.679 ms
84 bytes from 192.168.10.20 icmp_seq=41 ttl=64 time=0.637 ms
84 bytes from 192.168.10.20 icmp_seq=42 ttl=64 time=0.693 ms
84 bytes from 192.168.10.20 icmp_seq=43 ttl=64 time=0.630 ms

```

Failover Test Again:

```

RouterA(config)#interface g1/0
RouterA(config-if)#shutdown
RouterA(config-if)#
*Jun 13 23:33:17.211: %TRACKING-5-STATE: 1 interface Gi1/0 line-protocol Up->Down
*Jun 13 23:33:17.223: %HSRP-5-STATECHANGE: GigabitEthernet1/0 Grp 1 state Active -> Init
*Jun 13 23:33:19.211: %LINK-5-CHANGED: Interface GigabitEthernet1/0, changed state to administratively down
*Jun 13 23:33:20.211: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0, changed state to down

```

```

Jun 13 23:34:57.215: %HSRP-5-STATECHANGE: GigabitEthernet1/0 Grp 1 state Speak -> Standby
RouterB#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State   Active      Standby      Virtual IP
Gi1/0       1    100 P Active  local       unknown     192.168.10.1
RouterB#

```

Recovery Test

RouterA:

```

Jun 13 23:34:57.215: %HSRP-5-STATECHANGE: GigabitEthernet1/0 Grp 1 state Standby -> Speak
RouterA#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State   Active      Standby      Virtual IP
Gi1/0       1    120 P Active  local       192.168.10.3  192.168.10.1
RouterA#

```

RouterB:

```

Jun 13 23:34:57.215: %HSRP-5-STATECHANGE: GigabitEthernet1/0 Grp 1 state Speak -> Standby
RouterB#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State   Active      Standby      Virtual IP
Gi1/0       1    100 P Standby 192.168.10.2 local       192.168.10.1
RouterB#show ip route

```

Track Verification:

```

RouterA#show track
Track 1
  Interface GigabitEthernet1/0 line-protocol
  Line protocol is Up
    5 changes, last change 00:02:18
  Tracked by:
    HSRP GigabitEthernet1/0 1
RouterA#

```

Verify Routing Table:

```

RouterB#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
       C       192.168.10.0/24 is directly connected, GigabitEthernet1/0
       L       192.168.10.3/32 is directly connected, GigabitEthernet1/0
RouterB#show ip interface brief

```


Progress:

Week 3 objectives completed successfully and the project is ready for Week 4 integration and performance testing.